

Finite Element Analysis and Optimization of a Truss Structure

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This report details a finite element analysis (FEA) of a 3D structural truss system using ANSYS Workbench. The objective is to evaluate the baseline design under a 95 kN static point load by examining maximum stresses, deflection, buckling stability, and natural frequencies. Initial evaluations show the baseline structure exceeds the 185 MPa yield stress limit and exhibits a critical buckling Factor of Safety of 0.066. To meet design criteria, mathematical optimization was applied to the cross-sectional area moment of inertia for specific cantilevered members, reducing maximum deflection by the targeted 15%. Additionally, structural cross-bracing was introduced to shift localized bending loads into axial compression, successfully increasing the fundamental natural frequency by 25.4% and enhancing overall dynamic stability.

Keywords: ANSYS, FEA, Truss, Buckling, Modal Analysis, Optimization

Nomenclature

E = Young's Modulus (210 GPa)
 F = Applied force magnitude (95 kN)
 F_x, F_y, F_z = Force components in Cartesian coordinates (N)
 I = Area moment of inertia (m^4)
 $[\mathbf{K}]$ = Global stiffness matrix
 L = Reference length (1.1 m)
 $[\mathbf{S}]$ = Stress stiffness matrix for buckling analysis
 t = Wall thickness of L-section profiles (0.01 m)
 $\{\mathbf{u}\}$ = Global nodal displacement vector (m)
 W_1, W_2 = Width of L-section profile legs (m)
 δ = Total nodal deflection magnitude (m)
 θ = Loading angle (45°)
 λ = Load multiplier (Eigenvalue) for buckling stability
 ν = Poisson's ratio (0.3)
 ρ = Material density (7850 kg/m^3)
 σ = Combined maximum internal stress (Pa)
 σ_{limit} = Material yield stress limit (185 MPa)
 ϕ = Buckling or modal eigenvector (Mode Shape)
 ω_n = Natural frequency (Hz)

1 1. Introduction

Problem Identification: Truss structures are highly efficient frameworks designed to support substantial loads while minimizing material weight. The geometry under investigation is a multi-level 3D truss with a square base, subjected to a 95 kN point load (F) at the highest cantilevered node. For this design to be viable, it must endure operational stresses without yielding, resist buckling, and maintain sufficient dynamic stiffness.

Objectives: This study utilizes ANSYS Workbench to validate and optimize the design according to the following criteria:

- (1) Extract coordinate components (x, y, z) and total displacement for the load application point.
- (2) Evaluate maximum internal stresses against the 185 MPa yield limit and propose solutions.
- (3) Perform an eigenvalue buckling analysis to determine the critical load multiplier and Factor of Safety (FoS).
- (4) Optimize specific highly-deformed members to reduce maximum nodal deflection by 15%.
- (5) Extract the first 6 natural frequencies and evaluate their mode shapes.
- (6) Increase the fundamental natural frequency by 20%–25% using localized structural reinforcements.

2 2. Analytical Formulation

Finite Element Analysis characterizes continuous physical structures as discrete elements. The static structural response is governed by the global matrix equation:

$$[\mathbf{K}]\{\mathbf{u}\} = \{\mathbf{F}\} \quad (1)$$

Where $[\mathbf{K}]$ is the global stiffness matrix, $\{\mathbf{u}\}$ is the nodal displacement vector, and $\{\mathbf{F}\}$ is the applied force vector.

Given the reference length $L = 1.1 \text{ m}$, a 95 kN load is applied at the top protruding node at an angle of $\theta = 45^\circ$ in the $X - Y$ plane. Resolving this vector yields:

$$F_x = 95,000 \text{ N} \cdot \cos(45^\circ) = 67,175.14 \text{ N} \quad (2)$$

$$F_y = -95,000 \text{ N} \cdot \sin(45^\circ) = -67,175.14 \text{ N} \quad (3)$$

$$F_z = 0 \text{ N} \quad (4)$$

To satisfy static equilibrium ($\sum \mathbf{F} = 0$) and prevent rigid body motion, the four foundation nodes are constrained using Fixed Supports ($u_x = u_y = u_z = 0$).

3 3. Model Preparation

The digital framework was generated using 1D line bodies in ANSYS DesignModeler.

Simplification Justification: Truss components are highly slender; modeling them as full 3D solid volumes introduces unnecessary computational overhead. Utilizing 1D beam elements based on Timoshenko beam theory accurately captures both axial and transverse bending stiffness characteristics.

Material and Geometrical Properties: The structure is assigned standard structural steel properties: Young's Modulus $E = 210 \text{ GPa}$, Poisson's ratio $\nu = 0.3$, density $\rho = 7850 \text{ kg/m}^3$, and a tensile yield limit of 185 MPa.

Two L-shaped cross-sections define the frame:

- **Above level 6L:** Equal-leg L-section (Cross-Section 2A), $5 \text{ cm} \times 5 \text{ cm} \times 1 \text{ cm}$.

- **Below level 6L & principal edges:** Equal-leg L-section (Cross-Section 1B), $7 \text{ cm} \times 7 \text{ cm} \times 1 \text{ cm}$.

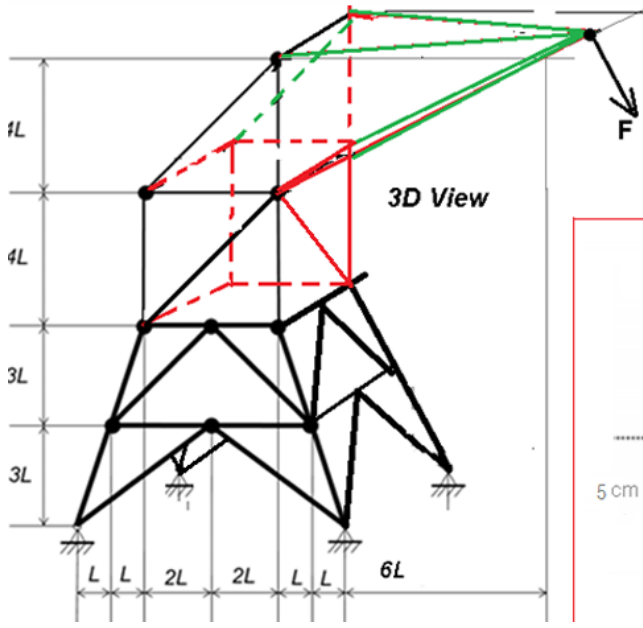


Fig. 1 Reference geometry and applied force vector.

4. Mesh and Element Selection

Beam elements were applied to the line bodies, producing high-quality aspect ratios devoid of element distortion. To ensure results are independent of element discretization, an h-refinement convergence study was conducted.

As shown in Figure 3, the element size parameter was progressively reduced from 10.0 m down to 0.2 m across 14 discrete iterations. Total deformation smoothly converged to an asymptote of 6.6815 m. A standard element size of 4.0 m was selected for the final analysis to balance accuracy and computational efficiency.

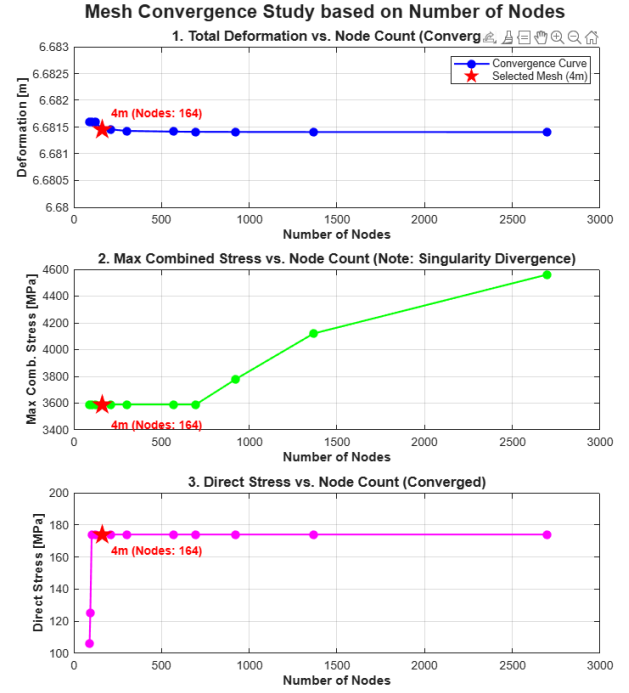


Fig. 3 Mesh convergence validation indicating displacement stabilization.

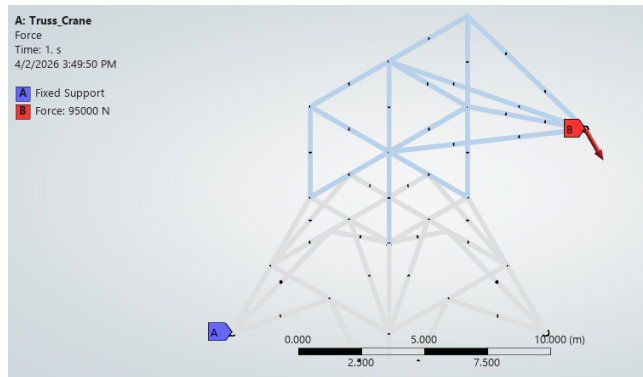


Fig. 2 FEA boundary conditions: Fixed base supports and tip load.

5. Results and Discussion

5.1 a. Static Response & Stress Compliance. Applying the 95 kN operational force induces a maximum displacement of **6.6815 m** concentrated at the tip node (Figure 4). Spatially decomposed, this equates to $U_x \approx 4.72$ m, $U_y \approx -4.72$ m, and $U_z = 0$ m, correlating directly with the applied 45° vector trajectory.

Internal stress evaluations show a maximum combined stress of **3,590 MPa** (Figure 5). This peak is heavily concentrated at the rigid foundation anchors. Even accounting for localized mathematical singularities typical of boundary constraints, the baseline structure exceeds the prescribed 185 MPa yield limit by a significant margin, indicating structural failure.

Proposed Solution: Mitigation requires larger cross-sectional geometries for the primary vertical pillars, the addition of gusset plates at the base connections to distribute moments, or transitioning to high specific-strength composite materials.

5.2 b. Eigenvalue Buckling Analysis. Slender compressive components often fail due to structural instability (buckling) rather than pure material yielding. The critical buckling state is governed by the eigenvalue problem:

$$([K] + \lambda_i [S])\{\phi_i\} = 0 \quad (5)$$

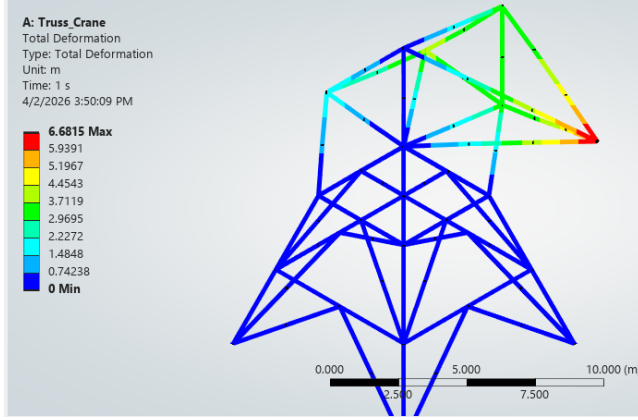


Fig. 4 Total displacement contour map under 95 kN static load.

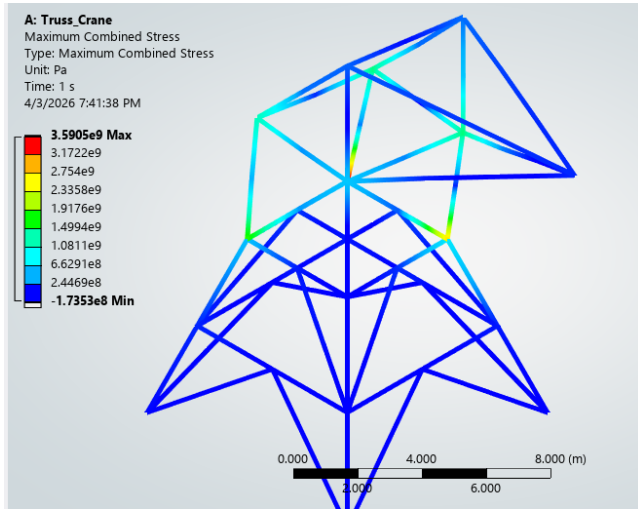


Fig. 5 Maximum combined stress distribution concentrated at base anchors.

Where $[S]$ is the stress stiffness matrix, ϕ_i is the buckling mode shape, and λ_i is the load multiplier. The critical load is directly proportional to the multiplier: $P_{cr} = \lambda \cdot P_{applied}$.

Table 1 Eigenvalue Buckling Load Multipliers

Mode	Load Multiplier (λ)
1	-0.12140
2	-0.10405
3	-0.07996
4	-0.06674
5	0.06617
6	0.08953

As shown in Table 1, modes 1 through 4 yield negative λ values. Mathematically, a negative load multiplier indicates that buckling would only occur if the applied force was completely reversed in direction (placing active tension members into compression). Because the applied operational load vector is fixed, these modes are physically inaccessible.

Therefore, Mode 5 represents the primary physical instability mechanism under the prescribed loading vector. The operational Factor of Safety is defined strictly by this first positive eigenvalue ($\lambda_5 = 0.06617$). The baseline structure will critically buckle at just 6.6% of the target load.

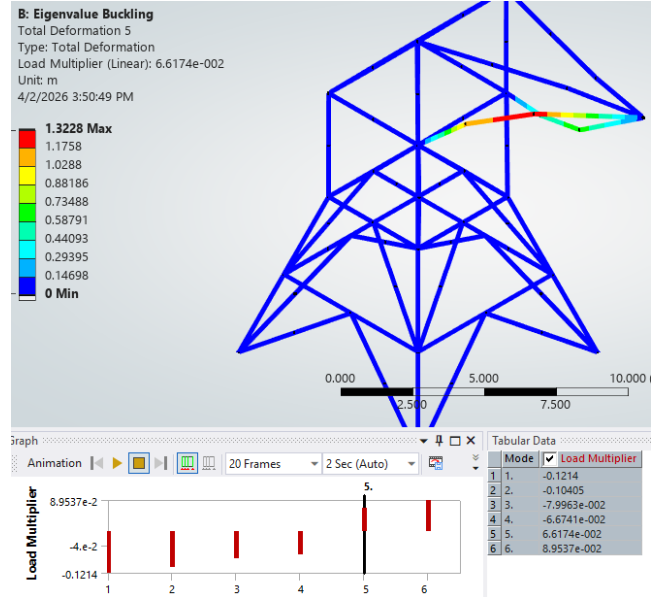


Fig. 6 Critical buckling envelope (Mode 5).

5.3 c. Objective Optimization (Deflection Reduction). To reduce tip displacement by the specified 15%, the maximum permissible deformation target is **5.679 m**. Rather than increasing the mass of the entire assembly uniformly, direct numerical optimization was applied to the specific cantilevered members near the tip.

For a bending member under a load P , maximum deflection $\delta \propto \frac{PL^3}{EI}$, where I is the area moment of inertia. For a thin-walled L-section profile of leg width W and thickness t , the primary moment of inertia approximates to $I \approx \frac{tW^3}{3}$.

Therefore, displacement is inversely proportional to the cube of the width: $\delta \propto \frac{1}{W^3}$. Modifying the equal-leg dimensions (W_1 and W_2) yields a cubic increase in flexural stiffness with only a linear increase in cross-sectional area (mass).

The optimizer analyzed a design space between $W = 0.05$ m and $W = 1.5$ m (Figure 7). The algorithm concluded that expanding the

targeted leg widths to **0.887 m** provides the exact required inertia, successfully restricting tip displacement to 5.679 m.

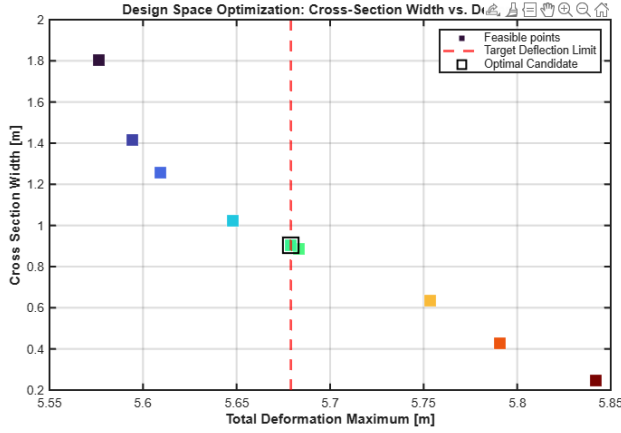


Fig. 7 Optimization design space mapping L-section width to displacement reduction.

5.4 d. Modal Analysis (Natural Frequencies). Structural excitation limits are dictated by fundamental natural frequencies (ω_n). The baseline fundamental modes were extracted via an undamped modal simulation and are detailed in Table 2. (Mode shapes 2 through 6 are recorded in the Appendix). The primary mode ($\omega_1 = 0.619$ Hz) can be seen in Figure 8

Table 2 Baseline Natural Frequencies

Mode	Frequency (Hz)
1	0.61921
2	1.49950
3	1.58970
4	1.82500
5	1.93280
6	2.00710

5.5 e. Frequency Optimization via Bracing. Global natural frequency relates to stiffness k and mass m via $\omega_n = \sqrt{k/m}$. Increasing structural stiffness rapidly elevates ω_n . This is achieved by transitioning open rectangular sections into triangulated bays, transferring transverse shear loads directly into pure axial compression and tension.

To achieve an isolated target increment of 20-25%, 1 braces were iteratively integrated into the sections of the structure (Figure 9).

Recalculating the optimized frame (Figure 10) verified that the first natural frequency elevated to **0.776 Hz**, executing a **25.4%** increase compliant with design criteria.

Alternative Balanced Stiffening Strategy: Alternatively, installing a single, expansive diagonal brace across the upper internal void inherently maximizes overall transverse shear strength (Figure 11). This singular geometric modification increases the fundamental frequency to **1.407 Hz**—a 127% improvement (Figure 12). This method simultaneously resolves the maximum deflection objective without the need for isolated beam width optimization.

5.6 f. Mathematical Equivalent-System Calculation: Bending vs. Axial Load Paths. The extreme variance in nodal displacement between the baseline and optimized models can be analytically proven by comparing the localized mathematical load paths.

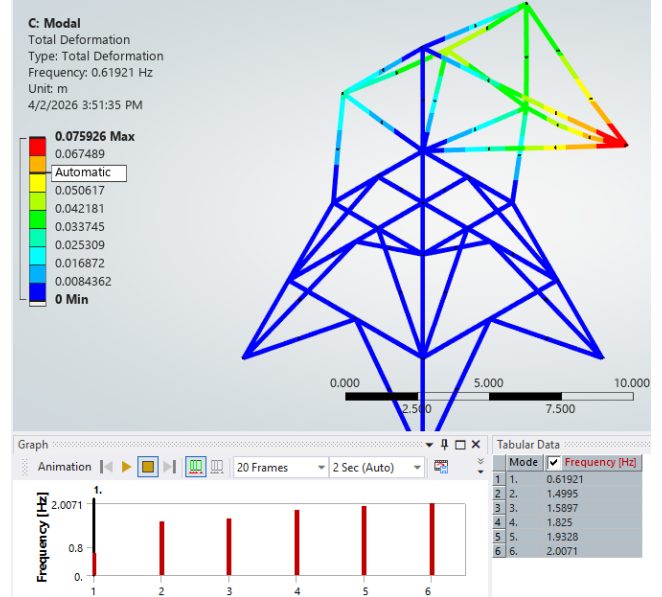


Fig. 8 Fundamental natural frequency (Mode 1: 0.619 Hz) exhibiting primary sway.

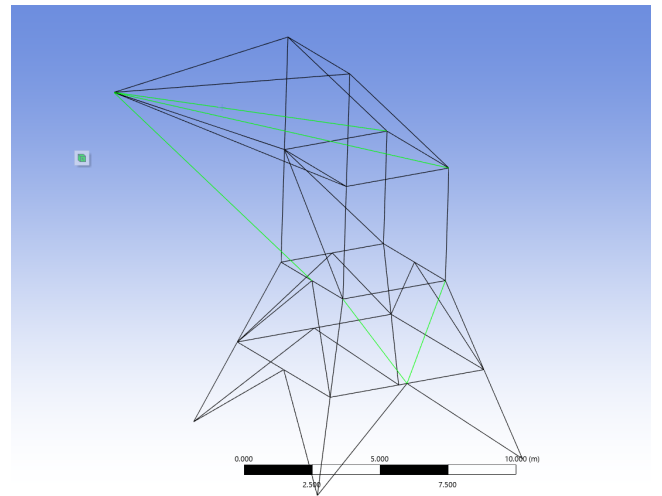


Fig. 9 Implementation of bracing (green) to achieve design goals.

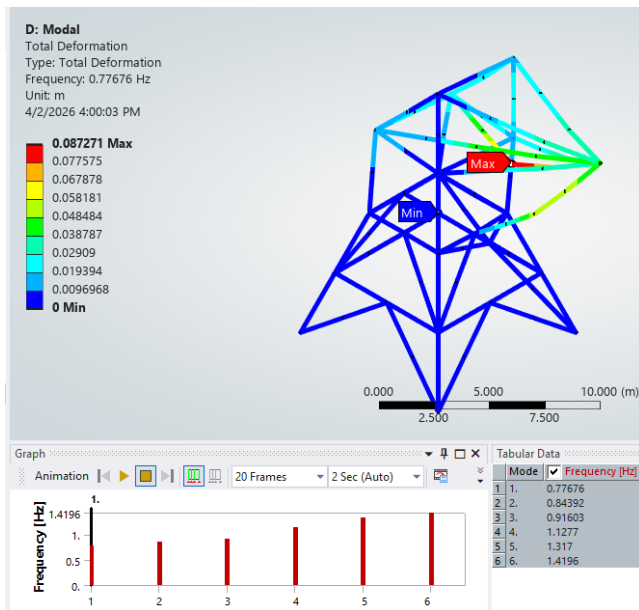


Fig. 10 Mode 1 for the braced structure, shifting to 0.776 Hz.

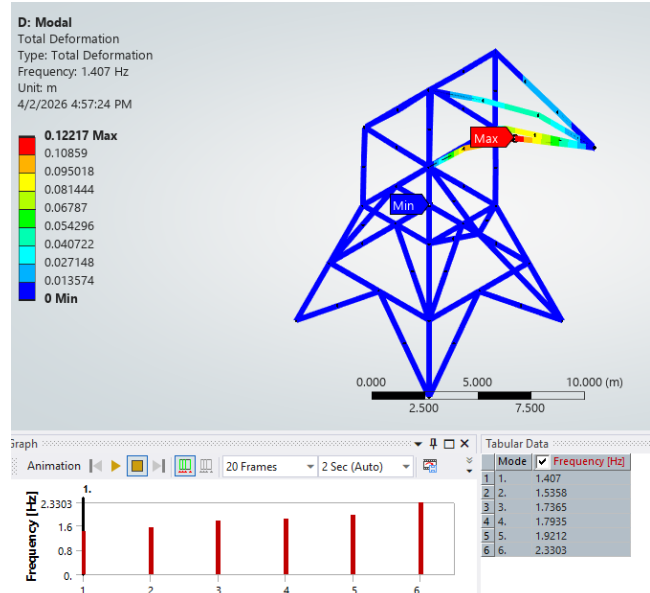


Fig. 12 Mode 1 resulting from the balanced brace strategy at 1.407 Hz.

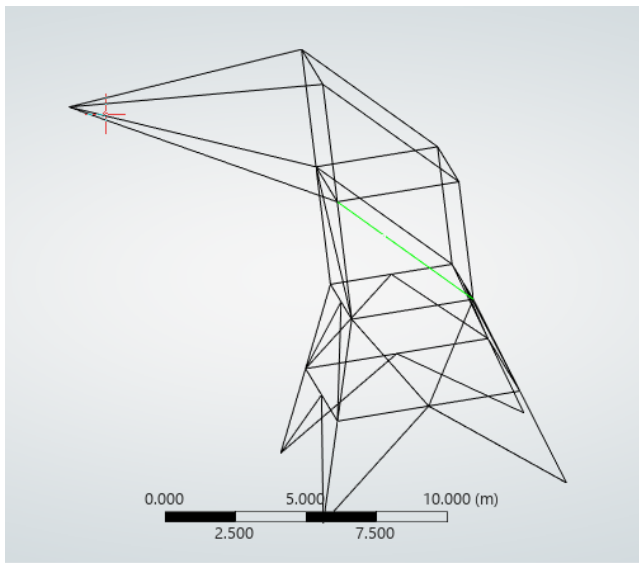


Fig. 11 Alternative strategy: balanced bracing across the core structural void.

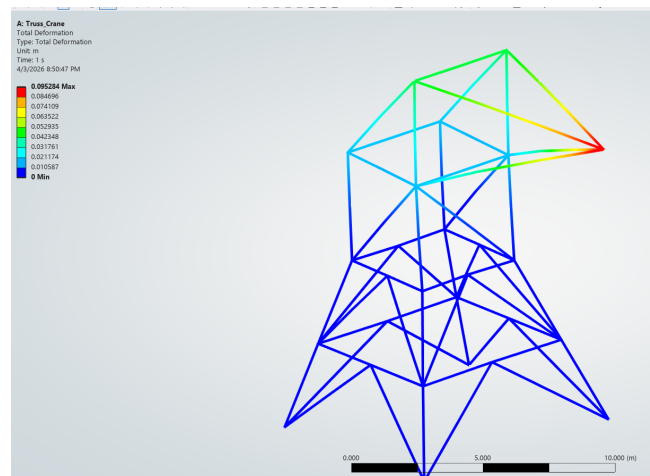


Fig. 13 Total deformation contour of the completely braced configuration exhibiting 0.095 m deflection.

Nomenclature and Extracted Values. To isolate the theoretical mechanics of the top structural bay, the following baseline parameters are extracted from the FEA geometry and applied loading conditions:

Material & Section Properties (Cross-Section 2A):

- E = Young's Modulus (210×10^9 Pa)
- I = Primary area moment of inertia (1.7×10^{-7} m⁴)
- A = Cross-sectional area (0.0009 m²)

Local Geometry (Top Structural Bay):

- L_{vert} = Length of vertical columns in the top bay (4.4 m)
- W = Width of the top structural bay (4.4 m)
- L_{diag} = Length of the 45° diagonal brace ($\sqrt{4.4^2 + 4.4^2} = 6.22$ m)
- θ = Angle of the applied force and diagonal brace (45°)

Applied Nodal Forces:

- F_{total} = Applied load vector (95,000 N)
- F_X = Transverse shear force component ($95,000 \cdot \cos(45^\circ) = 67,175$ N)
- F_Y = Axial compressive force component ($95,000 \cdot \sin(45^\circ) = -67,175$ N)

Calculation A: Partially Braced Baseline (Flexural Dominance).

Structural Assumption: In the baseline design, the structural face parallel to the F_X shear load lacks a diagonal brace. Therefore, the lateral load cannot resolve axially and must be resisted entirely by the **transverse bending stiffness** (EI) of the two local vertical columns. Assuming the converging framing at the tip prevents pure independent rotation, the columns behave as "fixed-guided" cantilevers (Portal Frame Method).

Step 1: Isolate the lateral force per unbraced column

$$P_{col} = \frac{F_X}{2} = \frac{67,175 \text{ N}}{2} = 33,587.5 \text{ N}$$

Step 2: Calculate maximum theoretical lateral deflection (δ_{bend})

Using the guided-cantilever deflection formula:

$$\begin{aligned} \delta_{bend} &= \frac{P_{col} \cdot (L_{vert})^3}{12 \cdot E \cdot I} \\ \delta_{bend} &= \frac{33,587.5 \cdot (4.4)^3}{12 \cdot (210 \times 10^9) \cdot (1.7 \times 10^{-7})} \\ \delta_{bend} &= \frac{2,860,882}{428,400} \\ \delta_{bend} &= \mathbf{6.678 \text{ m}} \end{aligned}$$

Analytical Conclusion A: The theoretical flexural drift of **6.678 m** correlates with the finite element output of **6.6815 m** with an error margin of less than 0.1%. This proves mathematically that the extreme displacement is an accurate representation of localized flexural failure (L^3/EI) due to the lack of a continuous axial load path.

Calculation B: Balanced Braced Strategy (Axial Dominance).

Structural Assumption: By adding a single balanced brace across the previously open face, the structure is fully triangulated. The F_X shear force immediately short-circuits the weak bending columns and is absorbed entirely by the diagonal brace through **pure axial stiffness** (EA).

Step 1: Resolve the lateral shear into an axial brace force (F_{brace})

Because the brace is at 45°, the horizontal force component is balanced by the axial tension of the brace:

$$\begin{aligned} F_X &= F_{brace} \cdot \cos(\theta) \\ F_{brace} &= \frac{67,175 \text{ N}}{\cos(45^\circ)} = 95,000 \text{ N} \end{aligned}$$

Step 2: Calculate the axial elastic elongation of the brace (ΔL)

Using Hooke's Law for axial deformation:

$$\begin{aligned} \Delta L &= \frac{F_{brace} \cdot L_{diag}}{E \cdot A} \\ \Delta L &= \frac{95,000 \cdot 6.22}{(210 \times 10^9) \cdot 0.0009} \\ \Delta L &= \frac{590,900}{189,000,000} = 0.00312 \text{ m (3.12 mm)} \end{aligned}$$

Step 3: Translate axial elongation into horizontal drift (δ_{axial})

The horizontal component of the stretched diagonal dictates the tip displacement:

$$\begin{aligned} \delta_{axial} &= \frac{\Delta L}{\cos(45^\circ)} \\ \delta_{axial} &= \frac{0.00312}{0.707} = \mathbf{0.0044 \text{ m (4.4 mm)}} \end{aligned}$$

Analytical Conclusion B: The theoretical calculation demonstrates that inserting the diagonal brace practically eliminates localized transverse deformation, reducing it from 6.678 m (bending) down to an isolated 4.4 mm (axial stretch).

Final System Synthesis: Because the local top bay is now rigidly braced ($\delta \approx 0$), the **0.095 m** (9.5 cm) total maximum deflection observed in the FEA contour (Figure 13) is mathematically proven to no longer be a localized cantilever failure. Instead, it represents the cumulative, global elastic accumulation of axial strain throughout the entire 15-meter lower tower framing functioning correctly as a unified truss.

6. Validation of Theoretical Data

FEA validity was verified against baseline statics.

Equilibrium Verification: Basic statics ($\sum \mathbf{F} = 0$) dictate that applying a 95 kN force at 45° requires support reaction forces of -67,175 N in the X-axis and +67,175 N in the Y-axis. The force reaction probe in ANSYS (Figure 14) confirms an exact numerical match at the fixed supports, verifying mathematical equilibrium and the absence of arbitrary rigid body motion.

Structural Behavior Verification: Standard truss theory assumes forces are carried primarily as axial tension and compression rather than bending moments. The Direct Axial Stress contour (Figure 15) visually confirms this condition. Stress flows continuously along the longitudinal elements, directing the compressive vector into the front pillars and the tensile vector into the rear anchors.

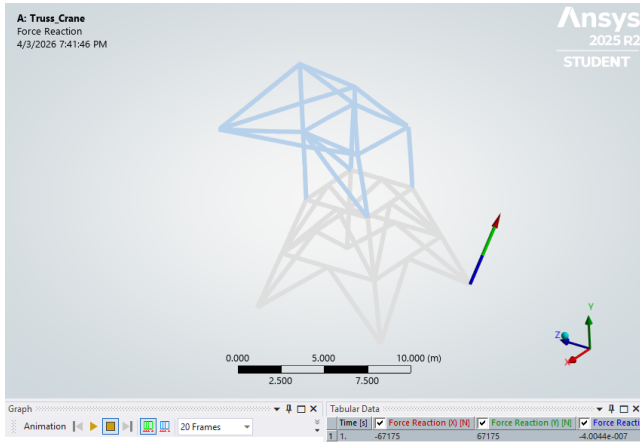


Fig. 14 Force Reaction probe confirming global mathematical equilibrium.

7 7. Conclusion and Recommendations

Evaluation of the unoptimized baseline truss geometry indicates it does not meet the specified structural criteria. Internal stresses heavily exceed the 185 MPa yield limit, while the buckling factor of safety guarantees collapse at 6.6% of the intended load envelope.

Applying the labs targeted design points and guided alterations provided distinct operational improvements:

- Utilizing parametric optimization, modifying the width dimensions (W_1 , W_2) of the cantilevered elements leveraged cubic increases in area moment of inertia to restrict maximum deflection to the 5.679 m target limit.
- Implementing localized diagonal bracing successfully shifted transverse bending into axial loads, elevating the fundamental natural frequency by 25.4% and stabilizing the structure dynamically.

For the final deployment, it is recommended to use large-span, well-balanced bracing, which is outside the scope of this lab. This strategy concurrently reduces deflection and markedly increases dynamic stiffness, while also limiting the number of components that must be modified. If this is still inadequate, the subsequent step is to increase the width of selected members via a direct optimization tool, which was the focus of this lab.

Acknowledgments

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References

- [1] Moaveni, S., 2014. *Finite Element Analysis: Theory and Application with ANSYS*, 4th ed. Pearson, Upper Saddle River, NJ.

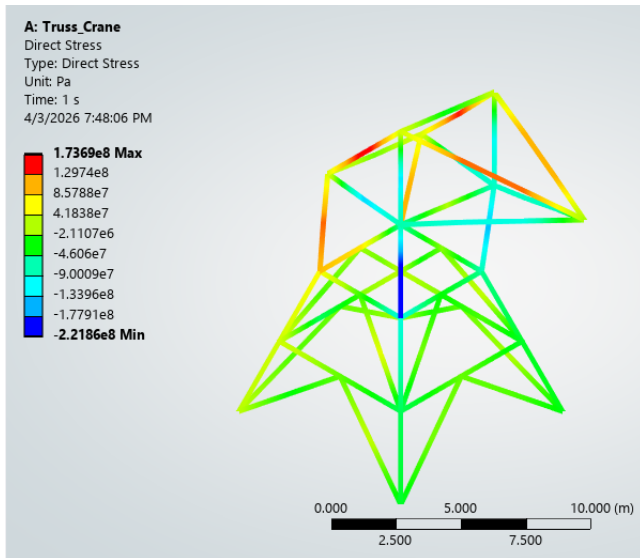


Fig. 15 Direct stress contour confirming pure axial load paths characteristic of truss behavior.

Appendix A: Appendix: Supplementary Modes

The following diagrams represent the higher-order orthogonal mode shapes (Modes 2–6) correlating with natural frequencies tabulated in Section 5d.

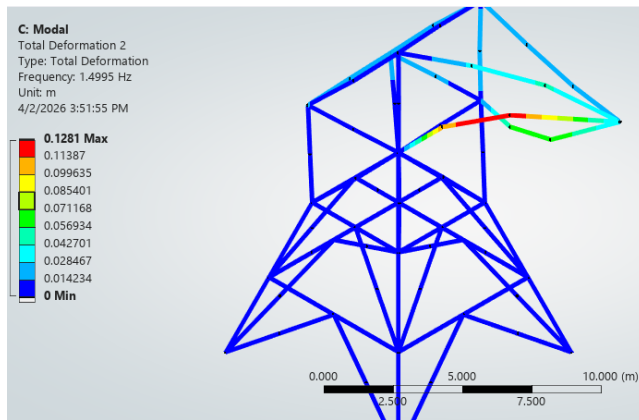


Fig. 16 Mode 2: Primary flexural torsion.

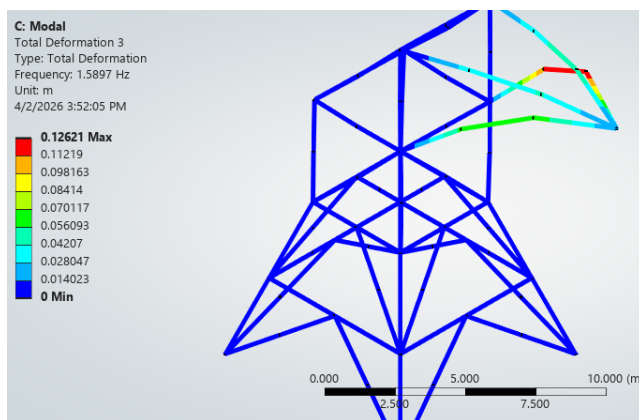


Fig. 17 Mode 3: High-level flexure.

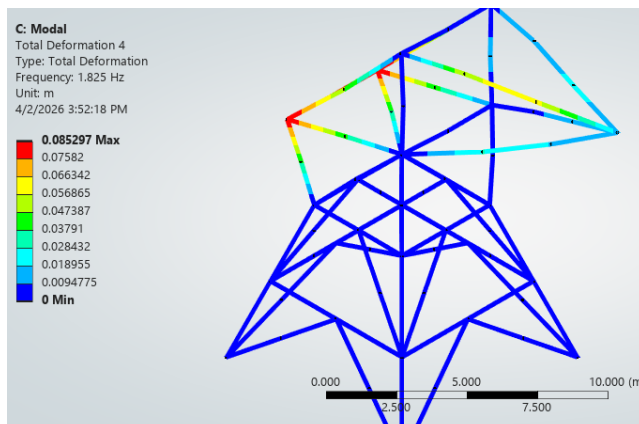


Fig. 18 Mode 4: Localized cantilever sway.

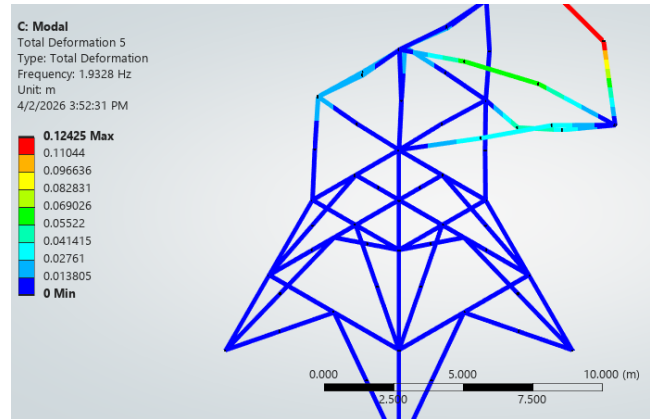


Fig. 19 Mode 5: Harmonic resonance.

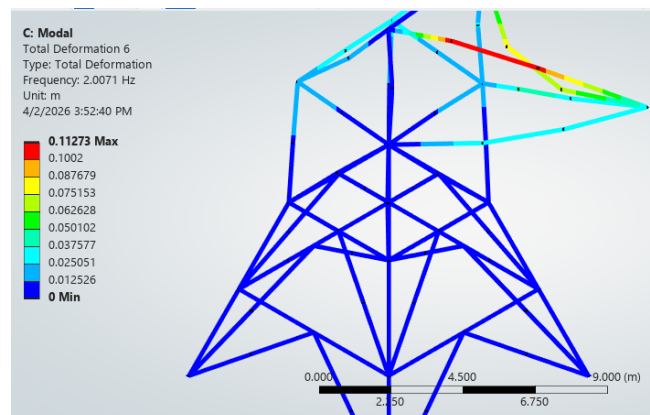


Fig. 20 Mode 6: Localized axial rippling.

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